# Latent.pk — Backtest Report

All 65 individual models + all 13 ensemble methods (A→M) with their model composition + comprehensive backtest stats.

**Generated:** 2026-10-23T20:58:24.076978 | **Trainable stocks:** 673 (60+ bars with volume>0) | **DB stocks:** 908 | **Models:** 65 | **Ensembles:** 13

## Section 1 — Ensemble Backtest Results (13 methods, ranked by win rate)

Each ensemble is a combination of one or more models. The 'Models in Ensemble' column lists all models that contribute to each ensemble's signal.

| # | Ensemble | Models in Ensemble | Trades | Win% | PF | Sharpe | ★ |
|---|---|---|---|---|---|---|---|
| 1 | F. CL-2 Ensemble (7 classifiers) | LightGBM-A (CL-2), LightGBM-B (CL-2), HistGBM (CL-2), ExtraTrees (CL-2), RandomForest (CL-2), Logistic (CL-2), MLP (CL-2) | 705 | 78.3 | 9.37 | 13.50 | ★★★★★ |
| 2 | G. PI-Ensemble (4 models) | Linear Regression (PI), Logistic Regression (PI), Pattern Matcher KNN (PI), Neural Network (PI) | 705 | 78.3 | 10.13 | 13.88 | ★★★★★ |
| 3 | J. X-Ensemble (10-strategy meta) | Trend Multi-TF (X), Breakout RelVol (X), Trend Pullback (X), RSI Divergence (X), BB Mean Reversion (X), Vol Contraction (X), Relative Strength (X), Market Regime (X), Meta-Label CG (X), Triple Barrier Dyn (X) | 705 | 78.3 | 4.03 | 8.13 | ★★★★★ |
| 4 | H. GM-Model (4-Pillar) | Technical Engine (GM), Announcements Engine (GM), Fundamental Engine (GM), Macro Engine (GM) | 705 | 57.3 | 1.38 | 1.73 | ★★★★· |
| 5 | A. Majority Vote | ALL 56 trained models (each model votes BUY/SELL/HOLD; majority wins) | 674 | 41.1 | 0.91 | -0.45 | ★★··· |
| 6 | B. Probability-Weighted | ALL 56 trained models (weighted by avgReturn) | 674 | 41.1 | 0.91 | -0.45 | ★★··· |
| 7 | C. Tier-Weighted | ALL 56 trained models (weighted by tier) | 674 | 41.1 | 0.91 | -0.45 | ★★··· |
| 8 | D. CL-Ensemble (LGB70+Log30) | LightGBM (CL), Logistic (CL-2) | 705 | 39.3 | 0.87 | -0.48 | ★★··· |
| 9 | E. CL-Extender (8-model weighted) | ARIMA, Holt-Winters, GARCH, Ridge Regression, Prophet, Kalman Filter, XGBoost (CL), LightGBM (CL), Random Forest (CL) | 705 | 39.3 | 0.87 | -0.48 | ★★··· |
| 10 | I. CG-Ensemble (swing-trader) | LightGBM (CG), XGBoost (CG) | 705 | 30.2 | 1.20 | 0.60 | ★···· |
| 11 | K. Gian-Ensemble (S/R) | Gian-Ensemble (GIAN) — rule-based (square-root S/R) | 705 | 28.1 | 1.19 | 0.59 | ★···· |

| # | Ensemble | Models in Ensemble | Trades | Win% | PF | Sharpe | ★ |
|---|---|---|---|---|---|---|---|
| 12 | L. Gian-Extender (GANN grid) | Gian-Extender (GIAN-EXT) — rule-based (GANN square-of-9) | 705 | 28.1 | 1.19 | 0.59 | ★···· |
| 13 | M. M-Ensemble (price-touch S/R) | M-Ensemble (M) — rule-based (historical price-touch S/R) | 705 | 28.1 | 1.19 | 0.59 | ★···· |

# Section 2 — Ensemble Composition + Detailed Stats

Per-ensemble breakdown: description, the models that compose it, and the backtest stats.

## A. Majority Vote

**Description:** Each model casts a vote (BUY/SELL/HOLD). The signal with the most votes wins.

**Models in this ensemble:** ALL 56 trained models (each model votes BUY/SELL/HOLD; majority wins)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 674 | 41.1% | -0.24% | 0.91 | -0.45 | -0.61 | 293.17% | ★★··· |

## B. Probability-Weighted

**Description:** Weighted by avgReturn across all models. Signal = BUY if avg > 0.01, SELL if < -0.01.

**Models in this ensemble:** ALL 56 trained models (weighted by avgReturn)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 674 | 41.1% | -0.24% | 0.91 | -0.45 | -0.61 | 293.17% | ★★··· |

## C. Tier-Weighted

**Description:** Weighted by tier (TIER_1 0.30, TIER_2 0.20, TIER_3 0.25, TIER_4 0.10, CL 0.15, CL_2 0.20, PI 0.20, GM 0.20, CG 0.20, X 0.20, GIAN 0.20).

**Models in this ensemble:** ALL 56 trained models (weighted by tier)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 674 | 41.1% | -0.24% | 0.91 | -0.45 | -0.61 | 293.17% | ★★··· |

## D. CL-Ensemble (LGB70+Log30)

**Description:** LightGBM (CL) 70% weight + Logistic (CL-2) 30% weight. Calibrated classifier ensemble.

**Models in this ensemble:** LightGBM (CL), Logistic (CL-2)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 39.3% | -0.51% | 0.87 | -0.48 | -0.63 | 971.02% | ★★··· |

## E. CL-Extender (8-model weighted)

**Description:** 8-model weighted ensemble from the CL tier (classical/linear models).

**Models in this ensemble:** ARIMA, Holt-Winters, GARCH, Ridge Regression, Prophet, Kalman Filter, XGBoost (CL), LightGBM (CL), Random Forest (CL)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 39.3% | -0.51% | 0.87 | -0.48 | -0.63 | 971.02% | ★★··· |

## F. CL-2 Ensemble (7 classifiers)

**Description:** 7 calibrated classifiers from the CL-2 tier. Triple-barrier target (TP1-before-SL in 5 days).

**Models in this ensemble:** LightGBM-A (CL-2), LightGBM-B (CL-2), HistGBM (CL-2), ExtraTrees (CL-2), RandomForest (CL-2), Logistic (CL-2), MLP (CL-2)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 78.3% | 35.42% | 9.37 | 13.5 | 19.56 | 383.61% | ★★★★★ |

## G. PI-Ensemble (4 models)

**Description:** 4 pattern & indicator models from the PI tier. Same support/ATR SL rule + TP1-before-SL target.

**Models in this ensemble:** Linear Regression (PI), Logistic Regression (PI), Pattern Matcher KNN (PI), Neural Network (PI)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 78.3% | 38.6% | 10.13 | 13.88 | 21.32 | 383.61% | ★★★★★ |

## H. GM-Model (4-Pillar)

**Description:** 4-Pillar Weighted Framework: Technical 35% + Announcements 30% + Fundamental 20% + Macro 15%.

**Models in this ensemble:** Technical Engine (GM), Announcements Engine (GM), Fundamental Engine (GM), Macro Engine (GM)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 57.3% | 2.85% | 1.38 | 1.73 | 2.08 | 388.75% | ★★★★· |

## I. CG-Ensemble (swing-trader)

**Description:** Probability averaging of 2 swing-trading classifiers + decile calibration. Triple-barrier, 10-day horizon, TP1=1R.

**Models in this ensemble:** LightGBM (CG), XGBoost (CG)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 30.2% | 0.83% | 1.2 | 0.6 | 0.47 | 383.61% | ★···· |

## J. X-Ensemble (10-strategy meta)

**Description:** 10 complementary strategies + LightGBM meta-labeling. 3-way split, purge+embargo, Platt+isotonic calibration.

**Models in this ensemble:** Trend Multi-TF (X), Breakout RelVol (X), Trend Pullback (X), RSI Divergence (X), BB Mean Reversion (X), Vol Contraction (X), Relative Strength (X), Market Regime (X), Meta-Label CG (X), Triple Barrier Dyn (X)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 78.3% | 12.75% | 4.03 | 8.13 | 7.03 | 383.61% | ★★★★★ |

## K. Gian-Ensemble (S/R)

**Description:** Square-root support/resistance with odd-number rule. BUY near S1/S2/S3, SELL near R1/R2/R3.

**Models in this ensemble:** Gian-Ensemble (GIAN) — rule-based (square-root S/R)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 28.1% | 0.81% | 1.19 | 0.59 | 0.45 | 383.61% | ★···· |

## L. Gian-Extender (GANN grid)

**Description:** Fixed GANN square-of-9 levels (1, 9, 25, 49, 81, 121, 169, 225, 289, 361, ...). BUY below, SELL above.

**Models in this ensemble:** Gian-Extender (GIAN-EXT) — rule-based (GANN square-of-9)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 28.1% | 0.81% | 1.19 | 0.59 | 0.45 | 383.61% | ★···· |

## M. M-Ensemble (price-touch S/R)

**Description:** Historical price-touch S/R with ATR clustering. Scan all highs/lows, cluster within 1×ATR, pick top 3.

**Models in this ensemble:** M-Ensemble (M) — rule-based (historical price-touch S/R)

| Trades | Win Rate | Avg Return | Profit Factor | Sharpe | Sortino | Max DD | Rating |
|---|---|---|---|---|---|---|---|
| 705 | 28.1% | 0.81% | 1.19 | 0.59 | 0.45 | 383.61% | ★···· |

## Section 3 — Individual Model Backtest Results (65 models, ranked by win rate)

All 65 models trained on all 673 trainable stocks (60+ bars with volume>0). Models that fail to train on a specific stock get a technical fallback signal (EMA/RSI/ATR) so every stock has all 65 model predictions.

| # | Model | Trades | Win% | Avg Ret% | PF | Sharpe | Sortino | Max DD% | ★ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | GARCH | 673 | 100.0 | 3.01 | 99.00 | 18.45 | 0.00 | 0.00 | ★★★★★ |
| 2 | Technical Engine (GM) | 674 | 81.5 | 1.00 | 5.56 | 6.34 | 2.21 | 57.10 | ★★★★★ |
| 3 | Prophet | 673 | 66.6 | 2.19 | 2.02 | 2.84 | 4.14 | 117.05 | ★★★★★ |
| 4 | Random Forest | 674 | 56.5 | 1.97 | 2.49 | 3.21 | 8.54 | 43.23 | ★★★★· |
| 5 | XGBoost | 674 | 56.4 | 3.80 | 3.30 | 1.81 | 12.94 | 57.34 | ★★★★· |
| 6 | CatBoost | 674 | 56.1 | 3.01 | 3.21 | 1.83 | 11.52 | 52.36 | ★★★★· |
| 7 | Random Forest (CL) | 673 | 55.4 | 3.59 | 3.32 | 1.62 | 11.64 | 67.65 | ★★★★· |
| 8 | LightGBM | 674 | 55.2 | 2.58 | 2.25 | 2.73 | 8.15 | 73.17 | ★★★★· |
| 9 | ARIMA | 673 | 54.8 | 0.14 | 1.66 | 1.78 | 2.40 | 10.25 | ★★★★· |
| 10 | Moirai (Salesforce) | 674 | 53.6 | 5.76 | 1.59 | 1.10 | 1.88 | 1355.28 | ★★★★· |
| 11 | LightGBM (CL) | 673 | 53.3 | 2.93 | 2.25 | 2.67 | 8.02 | 123.75 | ★★★★· |
| 12 | XGBoost (CL) | 673 | 53.3 | 5.33 | 3.02 | 1.51 | 13.59 | 109.57 | ★★★★· |
| 13 | LSTM | 674 | 48.7 | 0.63 | 1.15 | 0.57 | 0.60 | 278.29 | ★★★·· |
| 14 | CNN-LSTM | 674 | 48.1 | 0.62 | 1.15 | 0.57 | 0.60 | 273.33 | ★★★·· |
| 15 | XGBoost (CG) | 673 | 48.0 | -1.44 | 0.91 | -0.68 | -1.48 | 1735.14 | ★★★·· |
| 16 | GRU | 673 | 47.8 | 0.64 | 1.14 | 0.57 | 0.61 | 284.83 | ★★★·· |
| 17 | Lag-Llama | 674 | 47.8 | 0.63 | 1.14 | 0.56 | 0.61 | 284.83 | ★★★·· |
| 18 | Transformer-LSTM | 674 | 47.8 | 0.62 | 1.15 | 0.57 | 0.60 | 268.46 | ★★★·· |
| 19 | Ridge Regression | 674 | 47.0 | 0.15 | 1.03 | 0.12 | 0.15 | 403.68 | ★★★·· |
| 20 | Linear Regression (PI) | 673 | 45.0 | 0.08 | 1.02 | 0.08 | 0.10 | 456.55 | ★★★·· |
| 21 | Relative Strength (X) | 673 | 45.0 | 20.09 | 6.73 | 7.44 | 14.12 | 244.87 | ★★★·· |
| 22 | LightGBM (CG) | 673 | 44.1 | -2.92 | 0.78 | -1.66 | -3.26 | 1995.20 | ★★··· |
| 23 | Holt-Winters | 673 | 42.2 | -0.08 | 0.77 | -0.76 | -0.79 | 64.34 | ★★··· |

| # | Model | Trades | Win% | Avg Ret% | PF | Sharpe | Sortino | Max DD% | ★ |
|---|---|---|---|---|---|---|---|---|---|
| 24 | Chronos (Amazon) | 674 | 39.3 | -0.50 | 0.91 | -0.43 | -0.52 | 933.65 | ★★··· |
| 25 | Time-LLM | 674 | 39.3 | -0.75 | 0.87 | -0.64 | -0.78 | 1103.64 | ★★··· |
| 26 | Informer | 674 | 38.6 | -0.82 | 0.86 | -0.71 | -0.86 | 1153.03 | ★★··· |
| 27 | N-BEATS | 674 | 38.6 | -0.82 | 0.86 | -0.71 | -0.86 | 1153.03 | ★★··· |
| 28 | PatchTST | 674 | 38.6 | -0.82 | 0.86 | -0.71 | -0.86 | 1153.03 | ★★··· |
| 29 | TimesNet | 674 | 38.6 | -0.82 | 0.86 | -0.71 | -0.86 | 1153.03 | ★★··· |
| 30 | LightGBM-B (CL-2) | 673 | 38.3 | -5.37 | 0.57 | -3.73 | -6.77 | 3632.23 | ★★··· |
| 31 | Kalman Filter | 673 | 36.7 | -3.11 | 0.56 | -2.55 | -2.80 | 2286.95 | ★★··· |
| 32 | HistGBM (CL-2) | 673 | 36.4 | -5.80 | 0.53 | -4.12 | -6.98 | 3919.10 | ★★··· |
| 33 | LightGBM-A (CL-2) | 673 | 35.8 | -5.99 | 0.52 | -4.28 | -7.38 | 4044.85 | ★★··· |
| 34 | RandomForest (CL-2) | 673 | 31.8 | -3.88 | 0.43 | -5.01 | -7.85 | 2622.53 | ★···· |
| 35 | Triple Barrier Dyn (X) | 673 | 31.8 | 6.87 | 2.96 | 3.69 | 4.83 | 244.87 | ★···· |
| 36 | MLP (CL-2) | 673 | 31.4 | -4.59 | 0.47 | -4.18 | -5.84 | 3145.36 | ★···· |
| 37 | Trend Pullback (X) | 673 | 31.4 | 6.42 | 2.83 | 3.53 | 4.51 | 244.87 | ★···· |
| 38 | Pattern Matcher KNN (PI) | 673 | 31.1 | -16.62 | 0.23 | -8.51 | -10.75 | 11716.72 | ★···· |
| 39 | Vol Contraction (X) | 673 | 30.5 | 5.53 | 2.58 | 3.20 | 3.89 | 244.87 | ★···· |
| 40 | Announcements Engine (GM) | 674 | 30.4 | -0.58 | 0.91 | -0.41 | -0.78 | 1032.09 | ★···· |
| 41 | Trend Multi-TF (X) | 673 | 29.9 | 4.94 | 2.41 | 2.96 | 3.47 | 244.87 | ★···· |
| 42 | Macro Engine (GM) | 674 | 27.9 | -2.16 | 0.79 | -1.04 | -2.26 | 2478.96 | ★···· |
| 43 | Market Regime (X) | 673 | 27.3 | 2.41 | 1.69 | 1.77 | 1.69 | 244.87 | ★···· |
| 44 | Breakout RelVol (X) | 673 | 27.2 | 2.26 | 1.65 | 1.69 | 1.59 | 244.87 | ★···· |
| 45 | Logistic (CL-2) | 673 | 26.9 | -5.65 | 0.39 | -5.34 | -8.15 | 3797.30 | ★···· |
| 46 | Meta-Label CG (X) | 673 | 26.7 | 1.82 | 1.52 | 1.42 | 1.28 | 244.87 | ★···· |
| 47 | RSI Divergence (X) | 673 | 26.4 | 1.52 | 1.43 | 1.24 | 1.07 | 244.87 | ★···· |
| 48 | Logistic Regression (PI) | 673 | 26.0 | -14.75 | 0.22 | -9.60 | -14.21 | 10509.00 | ★···· |
| 49 | BB Mean Reversion (X) | 673 | 25.7 | 0.78 | 1.22 | 0.70 | 0.55 | 244.87 | ★···· |

| # | Model | Trades | Win% | Avg Ret% | PF | Sharpe | Sortino | Max DD% | ★ |
|---|---|---|---|---|---|---|---|---|---|
| 50 | Neural Network (PI) | 673 | 25.7 | -21.48 | 0.16 | -11.50 | -16.45 | 15021.16 | ★···· |
| 51 | ExtraTrees (CL-2) | 673 | 22.6 | -3.89 | 0.31 | -6.23 | -9.11 | 2617.45 | ····· |
| 52 | Fundamental Engine (GM) | 674 | 4.2 | -3.37 | 0.02 | -15.56 | -16.08 | 2271.05 | ····· |
| 53 | BB Mean Reversion | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 54 | Breakout RelVol | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 55 | Gian-Ensemble (GIAN) | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 56 | Gian-Extender (GIAN-EXT) | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 57 | M-Ensemble (M) | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 58 | Market Regime | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 59 | Meta-Label CG | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 60 | RSI Divergence | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 61 | Relative Strength | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 62 | Trend Multi-TF | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 63 | Trend Pullback | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 64 | Triple Barrier Dyn | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |
| 65 | Vol Contraction | 673 | 0.0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | ····· |